FPE Data Dictionary v3.0

Unified Master Finance-First F&B Financial Projection Engine

Version 3.0 | March 2026 | 73 Tables | 10 Layers + Optional Extension

73 Tables	10 Layers	6 Phases
Core + Optional	A through J	60-Week Roadmap

1. Introduction & Usage

This document is the authoritative Data Dictionary for the Unified Master v3 Finance-First F&B Financial Projection Engine. It is intended for database architects, backend engineers, and data analysts building or maintaining the projection engine.

The schema supports a planning-first architecture where financial projections are derived from high-level driver assumptions rather than granular transaction records. It supports multi-tenancy, scenario-based modeling, full version control, and end-to-end audit trails.

Every entry in this dictionary includes: table name, grain, scope classification, description, and a full field-level specification with data type, required flag, FK reference, description, and example value.

2. Schema Conventions

Convention	Rule	Example
Primary Keys	All tables use a UUID primary key named `id`. Generated server-side using gen_random_uuid().	id UUID DEFAULT gen_random_uuid()
Timestamps	All tables include `created_at` and `updated_at` as TIMESTAMPTZ.	created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
Audit Columns	All tables include `created_by` and `updated_by` (UUID → users).	created_by UUID REFERENCES users(id)
Multi-Tenancy	All tables include `company_id` (UUID) to isolate data per tenant.	company_id UUID NOT NULL REFERENCES companies(id)
Soft Delete	All tables include `deleted_at` (TIMESTAMPTZ, nullable). NULL = active record.	WHERE deleted_at IS NULL
Money Precision	DECIMAL(19,4) for all monetary amounts. Never use FLOAT for financial data.	amount DECIMAL(19,4) NOT NULL
Rate/Pct Fields	DECIMAL(5,4) for rates and percentages. 0.1500 represents 15%.	rate DECIMAL(5,4) -- 0.1500 = 15%
JSONB Metadata	JSONB `metadata` field on key tables for schema-agnostic extensibility.	metadata JSONB DEFAULT '{}'::jsonb
Naming	All table and column names use snake_case.	demand_drivers.growth_rate
FKs	All foreign keys use ON DELETE RESTRICT by default. Cascade deletes must be explicitly justified.	REFERENCES scenarios(id) ON DELETE RESTRICT
Indexes	Composite index on (company_id, scenario_id, period_id) on all fact tables. Partial index WHERE deleted_at IS NULL on all tables.	CREATE INDEX ... WHERE deleted_at IS NULL

3. Scope Taxonomy

Every table in this dictionary is classified into one of four scope levels. This drives the phased engineering delivery roadmap.

Scope Level	Target Phase	Description
Core v1	Phase 0-1 (Weeks 1-12)	Essential for MVP — minimum viable finance-first planning capability. All Core v1 tables must be delivered before go-live.
Recommended v1.5	Phase 2-3 (Weeks 13-36)	Critical additions for robust and auditable usage: governance, confidence scoring, basic risk, explainability.
Phase 2-3	Phase 3-5 (Weeks 25-60)	Advanced capabilities: portfolio optimization, full simulation suite, decision memory, Monte Carlo.
Optional Extension	Phase 6+ (Week 61+)	Operational detail not required for strategic planning. Includes raw order data, intraday demand, recipe BOM, shift logs.

4. Master Table Index — All 73 Tables

Layer	Table Name	Scope	Grain / Purpose
A	companies	Core v1	One tenant entity
A	planning_calendars	Core v1	One fiscal calendar config per company
A	planning_periods	Core v1	One time bucket per calendar
A	scenarios	Core v1	One named planning version (Base/Bull/Bear)
A	assumption_sets	Core v1	Versioned container of driver inputs per scenario
A	plan_versions	Core v1	Frozen/active plan snapshot
A	governance_events	Rec v1.5	Approval/lock/rejection audit events
A	assumption_lineage	Rec v1.5	Field-level change history for assumptions
B	countries	Core v1	Sovereign nation / legal jurisdiction
B	clusters	Core v1	Regional grouping (e.g. GCC)
B	macros	Core v1	City / metropolitan area
B	micros	Core v1	Neighborhood / delivery zone
B	kitchens	Core v1	Physical dark kitchen unit
B	platforms	Core v1	Sales channel / delivery aggregator
B	product_families	Core v1	Product category (Pizza, Sides, Beverages)
B	competitor_profiles	Phase 2-3	Competitor intelligence per market
C	demand_drivers	Core v1	Order volume projections per period/market
C	price_plans	Core v1	Pricing rules per product/platform/period
C	mix_plans	Core v1	Sales mix % per product family/platform
C	revenue_projection_lines	Core v1	Calculated revenue fact records
C	assumption_confidence	Rec v1.5	Confidence metadata per assumption record
D	unit_cost_profiles	Core v1	Per-unit COGS components per product
D	labor_models	Core v1	Staffing rules and pay rates per kitchen
D	headcount_plans	Core v1	Planned FTEs per kitchen/role/period
D	opex_plans	Core v1	Fixed/variable operating expense lines
D	marketing_plans	Core v1	Marketing spend by channel/period
D	capex_plans	Core v1	Capital expenditure schedule
D	working_capital_policies	Rec v1.5	Payment terms and inventory day policies
E	equity_rounds	Core v1	Equity fundraising events
E	debt_facilities	Core v1	Loan and debt instrument definitions
E	debt_repayment_schedules	Core v1	Amortization schedule per facility/period
E	capital_allocation_plans	Phase 2-3	Capital deployment rules per market
E	dilution_models	Rec v1.5	Cap table dilution calculations
F	pnl_projections	Core v1	Full projected P&L per period/kitchen
F	cashflow_projections	Core v1	Cash flow statement per period
F	balance_sheet_projections	Core v1	Balance sheet snapshot per period
F	unit_economics_projections	Core v1	Per-order economics metrics

G	kpi_projections	Core v1	Calculated KPI values per metric/period
G	driver_explainability	Rec v1.5	Driver contribution attribution
G	variance_attribution	Rec v1.5	Scenario-to-scenario variance bridge
G	performance_alerts	Rec v1.5	KPI threshold breach events
H	risk_objects	Rec v1.5	Risk factor catalog
H	risk_scenarios	Rec v1.5	Risk-adjusted scenario modifications
H	monte_carlo_runs	Phase 2-3	Monte Carlo simulation execution metadata
H	monte_carlo_summaries	Phase 2-3	Probabilistic result statistics per KPI
H	trigger_log	Rec v1.5	Planning engine trigger events fired
I	simulation_scenarios	Phase 2-3	Transient what-if simulation contexts
I	simulation_results	Phase 2-3	Transient KPI outputs from simulations
I	sensitivity_analyses	Phase 2-3	One-way sensitivity analysis results
I	decision_memory	Phase 2-3	Historical decision archive
I	decision_log	Phase 2-3	User action audit trail
J	portfolio_plans	Phase 2-3	Multi-market expansion strategy
J	market_portfolio_scores	Phase 2-3	Attractiveness/readiness scores per market
J	portfolio_capital_allocations	Phase 2-3	Capital budget distribution per market
J	portfolio_return_metrics	Phase 2-3	Aggregate portfolio IRR/NPV/MOIC
Opt	order_headers	Extension	Raw transaction order headers
Opt	order_lines	Extension	Raw order line items
Opt	kitchen_stations	Extension	Production station definitions
Opt	staff_roster	Extension	Actual staff shift records
Opt	actual_ingestion_log	Extension	Data import event log
Opt	intraday_demand_buckets	Extension	30-min/hourly demand buckets
Opt	operational_events	Extension	Kitchen incident log
Opt	product_variants	Extension	SKU-level product variants
Opt	modifier_types	Extension	Product add-on/modifier options
Opt	ingredient_bom	Extension	Bill of materials per variant
Opt	recipe_histories	Extension	Recipe version history

5. Layer A: Planning Spine & Governance

companies

Scope: Core v1 | Grain: One tenant entity

Root tenant entity. Every other table references this via company_id for data isolation in a multi-tenant deployment.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	550e8400-e29b-41d4...
name	VARCHAR(255)	Y		Legal company name	Pizza Co LLC
trade_name	VARCHAR(255)	N		Operating/brand name	Hot Slice
base_currency	CHAR(3)	Y		ISO 4217 reporting currency	AED
fiscal_year_start_month	INT	Y		Month number 1-12	1
country_code	CHAR(2)	Y		ISO 3166-1 alpha-2	AE
timezone	VARCHAR(50)	Y		IANA timezone identifier	Asia/Dubai
metadata	JSONB	N		Flexible extra attributes	{"tax_id":"TRN12345"}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

planning_calendars

Scope: Core v1 | Grain: One fiscal calendar configuration per company

Defines the fiscal calendar structure (monthly, quarterly) used to generate planning periods. Each company may have one active calendar.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
name	VARCHAR(100)	Y		Calendar label	Standard Monthly 2026
period_type	VARCHAR(20)	Y		MONTHLY, QUARTERLY, WEEKLY	MONTHLY
first_period_start	DATE	Y		Start date of first period	2026-01-01
last_period_end	DATE	Y		End date of last period	2028-12-31
is_active	BOOLEAN	Y		Only one active calendar per company	true
metadata	JSONB	N		Additional config	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04

updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

planning_periods

Scope: Core v1 | Grain: One time bucket (month/quarter) per calendar

Represents individual time buckets generated from the planning calendar. All driver and projection fact tables reference a planning_period_id.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
calendar_id	UUID	Y	planning_calendars.id	Parent calendar	...
period_label	VARCHAR(50)	Y		Human label	Jan 2026
start_date	DATE	Y		First day of period	2026-01-01
end_date	DATE	Y		Last day of period	2026-01-31
fiscal_year	INT	Y		Fiscal year integer	2026
fiscal_quarter	INT	Y		Quarter 1-4	1
fiscal_month	INT	Y		Month 1-12	1
is_closed	BOOLEAN	Y		Actuals locked if true	false
sequence_no	INT	Y		Ordered position in calendar	1
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

scenarios

Scope: Core v1 | Grain: One named planning version (Base, Bull, Bear, Stress)

Each scenario represents a distinct view of the future. Drivers and projections are always keyed to a scenario to allow parallel forecasts and scenario comparison.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
name	VARCHAR(100)	Y		Scenario label	Base Case 2026
type	VARCHAR(20)	Y		BASE|UPSIDE|DOWNSIDE|STRESS|BUDGET|FORECAST	BASE
description	TEXT	N		Narrative context	Conservative growth...
is_active	BOOLEAN	Y		Active flag	true
base_scenario_id	UUID	N	scenarios.id	Parent for cloned scenarios	...

is_locked	BOOLEAN	Y		Prevents further edits if true	false
owner_id	UUID	N	users.id	Responsible planner	...
approval_status	VARCHAR(20)	Y		DRAFT|PENDING|APPROVED|REJECTED	DRAFT
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

assumption_sets

Scope: Core v1 | Grain: One versioned container of driver inputs per scenario

Groups all driver assumptions into a named, versioned container. Each planning cycle should create a new assumption set rather than overwrite previous ones, preserving audit trails.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Parent scenario	...
name	VARCHAR(100)	Y		Set label	Q1 2026 Base Assumptions
version	VARCHAR(20)	Y		Semver or date label	v1.2
status	VARCHAR(20)	Y		DRAFT|ACTIVE|ARCHIVED	ACTIVE
confidence_level	VARCHAR(20)	N		HIGH|MEDIUM|LOW	MEDIUM
owner_id	UUID	N	users.id	Owner/analyst	...
approved_by	UUID	N	users.id	Approver user ID	...
approved_at	TIMESTAMPTZ	N		Approval timestamp	2026-01-10 09:00:00+04
notes	TEXT	N		Free-text commentary	Reviewed with CFO
metadata	JSONB	N		Extra attributes	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

plan_versions

Scope: Core v1 | Grain: One frozen or active snapshot of a plan

Records snapshots (frozen or active) of planning outputs. Enables budget-vs-forecast-vs-actual comparison and full version history.

Field Name	Type	Req	FK / Reference	Description	Example

id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Parent scenario	...
assumption_set_id	UUID	Y	assumption_sets.id	Linked assumption set	...
version_tag	VARCHAR(50)	Y		Label e.g. v1.0 or 2026-Q1	2026-Q1-Base
is_frozen	BOOLEAN	Y		Read-only if true	false
frozen_at	TIMESTAMPTZ	N		Timestamp of freeze	...
frozen_by	UUID	N	users.id	User who froze	...
label	VARCHAR(100)	N		Display label	Budget 2026
notes	TEXT	N		Commentary	Board approved
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

governance_events

Scope: Recommended v1.5 | Grain: One governance action (approval/lock/rejection)

Audit log for governance actions taken on plans, assumption sets, or scenarios. Supports approval workflows and regulatory compliance.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
event_type	VARCHAR(30)	Y		APPROVAL| REJECTION| LOCK|UNLOCK| COMMENT	APPROVAL
entity_type	VARCHAR(30)	Y		Type of governed entity	assumption_set
entity_id	UUID	Y		ID of the governed entity	...
actor_id	UUID	Y	users.id	User performing the action	...
reason	TEXT	N		Justification / comment	Approved for Q1 board
occurred_at	TIMESTAMPTZ	Y		When the event occurred	2026-01-08 10:00+04
metadata	JSONB	N		Additional context	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

assumption_lineage

Scope: Recommended v1.5 | Grain: One change-history entry per assumption field modification

Tracks every change made to driver assumption values, recording who changed what, when, and why. Supports full audit trail and regulatory compliance.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Parent assumption set	...
table_name	VARCHAR(100)	Y		Source table changed	demand_drivers
record_id	UUID	Y		ID of changed record	...
field_name	VARCHAR(100)	Y		Column that changed	growth_rate
old_value	TEXT	N		Value before change	0.05
new_value	TEXT	Y		Value after change	0.07
changed_by	UUID	Y	users.id	User making change	...
changed_at	TIMESTAMPTZ	Y		Timestamp of change	2026-01-12 11:00+04
change_reason	TEXT	N		Free-text justification	Revised upward per sales report
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

6. Layer B: Product & Market Reference

countries

Scope: Core v1 | Grain: One sovereign nation / legal jurisdiction

Top-level geographic hierarchy. Captures regulatory, economic, and operational parameters at country level.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
iso_alpha2	CHAR(2)	Y		ISO 3166-1 alpha-2 code	AE
iso_alpha3	CHAR(3)	Y		ISO 3166-1 alpha-3 code	ARE
name	VARCHAR(100)	Y		Country name	United Arab Emirates
currency_code	CHAR(3)	Y		ISO 4217 currency	AED
vat_rate	DECIMAL(5,4)	N		Standard VAT rate	0.0500
corporate_tax_rate	DECIMAL(5,4)	N		Standard corporate tax rate	0.0900
regulatory_notes	TEXT	N		Key regulatory considerations	Food license required from FSSAI
is_active	BOOLEAN	Y		Whether operating in this country	true
metadata	JSONB	N		Macro indices, ease of doing business	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

clusters

Scope: Core v1 | Grain: One regional cluster (e.g. GCC, Levant)

Groups countries into regional clusters for portfolio-level planning and regional rollup reporting.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
country_id	UUID	Y	countries.id	Parent country	...
name	VARCHAR(100)	Y		Cluster name	GCC
code	VARCHAR(20)	Y		Short identifier	GCC-01
description	TEXT	N		Description/ notes	Gulf Cooperation Council region
is_active	BOOLEAN	Y		Active flag	true
metadata	JSONB	N		Extra attributes	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04

created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

macros

Scope: Core v1 | Grain: One city or metropolitan area

City-level geographic entity. Captures city-specific market data including population, competition density, labor market, and rent index.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
cluster_id	UUID	Y	clusters.id	Parent cluster/region	...
name	VARCHAR(100)	Y		City name	Dubai
code	VARCHAR(20)	Y		Short identifier	DXB
population	INT	N		Estimated population	3600000
gdp_per_capita_usd	DECIMAL(15,2)	N		GDP per capita in USD	43000.00
avg_rent_per_sqft	DECIMAL(10,2)	N		Average commercial rent/sqft/yr	150.00
labor_cost_index	DECIMAL(5,4)	N		Relative labor cost index	1.0000
competition_density	DECIMAL(5,2)	N		Competitors per sqkm	3.50
is_active	BOOLEAN	Y		Active flag	true
metadata	JSONB	N		Other city data	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

micros

Scope: Core v1 | Grain: One neighborhood or delivery zone

Neighborhood-level entity representing a delivery catchment area. Used for kitchen placement, demand modeling, and attractiveness scoring.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
macro_id	UUID	Y	macros.id	Parent city	...
name	VARCHAR(100)	Y		Area name	JLT - Cluster A
code	VARCHAR(30)	Y		Short code	DXB-JLT-A
catchment_radius_km	DECIMAL(5,2)	N		Delivery radius in km	3.00
estimated_population	INT	N		Resident + daytime pop	85000
office_density_score	DECIMAL(3,2)	N		Office worker density (0-5)	4.20

food_delivery_index	DECIMAL(5,4)	N		Relative delivery demand index	1.2500
attractiveness_score	DECIMAL(5,2)	N		Composite attractiveness 0-100	74.50
launch_status	VARCHAR(20)	Y		PIPELINE| PLANNED| ACTIVE|PAUSED| CLOSED	PLANNED
metadata	JSONB	N		GIS coordinates, notes	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

kitchens

Scope: Core v1 | Grain: One physical dark kitchen unit

Represents a physical (or virtual) kitchen production unit. Links geographic location to financial and capacity planning.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
micro_id	UUID	Y	micros.id	Parent micro area	...
name	VARCHAR(100)	Y		Kitchen identifier	DXB-JLT-01
code	VARCHAR(30)	Y		Short code	K-001
status	VARCHAR(20)	Y		PLANNED| CONSTRUCTION| ACTIVE|CLOSED| PAUSED	ACTIVE
opening_date	DATE	N		Expected or actual opening date	2025-01-15
closing_date	DATE	N		If closed, closing date	
size_sqft	DECIMAL(10,2)	N		Total kitchen floor area	420.00
rent_per_annum	DECIMAL(19,4)	N		Annual base rent in AED	132000.0000
max_daily_capacity	INT	N		Max orders per day (theoretical)	400
kitchen_format	VARCHAR(30)	N		DARK_KITCHEN| SHARED| STANDALONE	DARK_KITCHEN
metadata	JSONB	N		Equipment list, notes	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...

deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

platforms

Scope: Core v1 | Grain: One sales channel or delivery aggregator

Represents a sales/distribution channel through which orders are received. Includes aggregators (Talabat, Deliveroo) and own-channel options.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
name	VARCHAR(100)	Y		Platform name	Talabat
type	VARCHAR(30)	Y		AGGREGATOR|DIRECT|B2B|DINE_IN	AGGREGATOR
commission_rate	DECIMAL(5,4)	N		Standard commission on GMV	0.2000
avg_payout_days	INT	N		Average payment settlement days	14
integration_method	VARCHAR(50)	N		API|TABLET|MANUAL	API
is_active	BOOLEAN	Y		Whether platform is in use	true
metadata	JSONB	N		Contractual notes, region coverage	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

product_families

Scope: Core v1 | Grain: One product category / family

High-level product grouping (e.g. Pizzas, Sides, Beverages, Desserts). Used as the primary planning grain for revenue, cost, and mix modeling.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
name	VARCHAR(100)	Y		Family name	Pizza
code	VARCHAR(20)	Y		Short code	PIZ
is_food	BOOLEAN	Y		Food item vs beverage/other	true
display_order	INT	N		Sort order for UI	1
parent_id	UUID	N	product_families.id	Parent family for hierarchy	
is_active	BOOLEAN	Y		Active flag	true
metadata	JSONB	N		Allergen info, prep complexity	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04

created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

competitor_profiles

Scope: Phase 2-3 | Grain: One competitor entity within a specific market

Captures competitive intelligence for market analysis and attractiveness scoring. Used in Phase 2-3 for competitive positioning and market response modeling.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
micro_id	UUID	Y	micros.id	Market micro area	...
name	VARCHAR(100)	Y		Competitor name	Pizza Hut JLT
type	VARCHAR(30)	N		DARK_KITCHEN| QSR| RESTAURANT| GHOST	QSR
estimated_daily_orders	INT	N		Estimated orders per day	250
avg_price_aed	DECIMAL(10,2)	N		Average order value in AED	65.00
market_share_pct	DECIMAL(5,4)	N		Estimated market share	0.1800
platform_presence	JSONB	N		Which platforms they are on	["Talabat","Deliveroo"]
threat_level	VARCHAR(20)	N		LOW|MEDIUM| HIGH|CRITICAL	MEDIUM
notes	TEXT	N		Intel notes	Recently relaunched menu
metadata	JSONB	N		Additional data points	{}
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

7. Layer C: Commercial Driver Engine

demand_drivers

Scope: Core v1 | Grain: One demand projection record per period / market / scenario

Core revenue driver capturing expected order volumes per planning period and market scope. Drives all revenue projection calculations.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Parent assumption set	...
period_id	UUID	Y	planning_periods.id	Planning period	...
scenario_id	UUID	Y	scenarios.id	Parent scenario	...
market_scope_type	VARCHAR(20)	N		MICRO|MACRO|CLUSTER|COUNTRY|ALL	MICRO
market_scope_id	UUID	N		FK to relevant geo table	...
kitchen_id	UUID	N	kitchens.id	Specific kitchen if applicable	...
platform_id	UUID	N	platforms.id	Platform filter if applicable	...
product_family_id	UUID	N	product_families.id	Product family filter	...
base_orders	DECIMAL(19,4)	Y		Baseline order volume for period	1500.0000
growth_rate	DECIMAL(5,4)	Y		Month-on-month growth rate	0.0500
seasonality_factor	DECIMAL(5,4)	Y		Seasonal index multiplier	1.1000
penetration_rate	DECIMAL(5,4)	N		Market penetration %	0.0320
marketing_uplift	DECIMAL(5,4)	N		Expected uplift from marketing	0.0800
platform_mix_pct	DECIMAL(5,4)	N		% of demand from this platform	0.4200
projected_orders	DECIMAL(19,4)	N		Computed: base * growth * seasonality	1732.5000
confidence_score	DECIMAL(3,2)	N		Driver confidence 0.00-1.00	0.75
notes	TEXT	N		Analyst commentary	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

price_plans

Scope: Core v1 | Grain: One pricing rule per product family / platform / period / scenario

Defines pricing assumptions including list price, discount rates, and net selling price per product family and platform combination.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Parent assumption set	...
period_id	UUID	Y	planning_periods.id	Planning period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
product_family_id	UUID	Y	product_families.id	Product family	...
platform_id	UUID	N	platforms.id	Platform (null = all)	...
list_price	DECIMAL(19,4)	Y		Gross listed price in base currency	45.0000
discount_rate	DECIMAL(5,4)	Y		Average discount percentage	0.1000
platform_commission	DECIMAL(5,4)	N		Platform commission override	0.2000
net_price	DECIMAL(19,4)	N		Computed net selling price	40.5000
price_elasticity	DECIMAL(5,4)	N		Own-price elasticity coefficient	-1.2000
notes	TEXT	N		Pricing rationale	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

mix_plans

Scope: Core v1 | Grain: One product/platform mix assumption per period / scenario

Captures the expected sales mix — what percentage of total orders belongs to each product family and platform. Used in blended AOV and margin calculations.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Parent assumption set	...
period_id	UUID	Y	planning_periods.id	Planning period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
product_family_id	UUID	Y	product_families.id	Product family	...
platform_id	UUID	N	platforms.id	Platform (null = all)	...
mix_pct	DECIMAL(5,4)	Y		% share of total orders	0.6500
aov_multiplier	DECIMAL(5,4)	N		AOV relative to baseline	1.0500
notes	TEXT	N		Commentary	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...

created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

revenue_projection_lines

Scope: Core v1 | Grain: One calculated revenue line per period / market / product / scenario

Stores computed revenue outputs derived from demand drivers, price plans, and mix plans. These are the primary revenue fact records feeding the P&L.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	Y	planning_periods.id	Period	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = rolled up)	...
platform_id	UUID	N	platforms.id	Platform (null = rolled up)	...
product_family_id	UUID	N	product_families.id	Product family (null = all)	...
projected_orders	DECIMAL(19,4)	Y		Total orders for this line	1732.5000
avg_order_value	DECIMAL(19,4)	Y		Blended AOV	52.5000
gross_revenue	DECIMAL(19,4)	Y		Orders * AOV	90956.2500
discounts	DECIMAL(19,4)	Y		Total discount value	9095.6250
platform_commissions	DECIMAL(19,4)	Y		Aggregator fees	18191.2500
net_revenue	DECIMAL(19,4)	Y		After discounts & commissions	63669.3750
confidence_score	DECIMAL(3,2)	N		Output confidence 0-1	0.78
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

assumption_confidence

Scope: Recommended v1.5 | Grain: One confidence metadata record per assumption record

Stores confidence scores, evidence sources, and volatility flags for individual driver assumption records. Enables explainability and risk tagging.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Parent set	...

table_name	VARCHAR(100)	Y		Source table (e.g. demand_drivers)	demand_drivers
record_id	UUID	Y		ID of the assumption record	...
confidence_level	VARCHAR(20)	Y		HIGH|MEDIUM|LOW|SPECULATIVE	MEDIUM
confidence_score	DECIMAL(3,2)	N		Numeric score 0.00-1.00	0.72
volatility_flag	BOOLEAN	N		True if high sensitivity to change	false
evidence_source	TEXT	N		Source data or rationale	Market survey Jan 2026
review_date	DATE	N		Next scheduled review	2026-04-01
notes	TEXT	N		Additional commentary	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

8. Layer D: Cost Driver Engine

unit_cost_profiles

Scope: Core v1 | Grain: One unit cost record per product family / period / scenario

Defines per-unit cost of goods sold (COGS) components including food cost, packaging, and waste factors. Key input for gross profit calculation.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Parent assumption set	...
period_id	UUID	Y	planning_periods.id	Period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
product_family_id	UUID	Y	product_families.id	Product family	...
food_cost_pct	DECIMAL(5,4)	Y		Food cost as % of net revenue	0.2500
packaging_cost_per_order	DECIMAL(19,4)	Y		Fixed packaging cost per order	1.5000
waste_factor_pct	DECIMAL(5,4)	N		Additional waste / yield loss %	0.0300
third_party_cost	DECIMAL(19,4)	N		Other variable unit costs	0.5000
total_cogs_per_order	DECIMAL(19,4)	N		Computed total COGS per order	13.0000
notes	TEXT	N		Cost assumptions rationale	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

labor_models

Scope: Core v1 | Grain: One labor model per kitchen / scenario / period

Defines staffing assumptions including roles, hours, and pay rates for each kitchen. Feeds into payroll cost projections.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Parent assumption set	...
period_id	UUID	Y	planning_periods.id	Period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = template)	...
role_name	VARCHAR(100)	Y		Role title	Pizza Chef
role_type	VARCHAR(20)	Y		KITCHEN| DELIVERY| MANAGEMENT| SUPPORT	KITCHEN

headcount	DECIMAL(5,2)	Y		FTE count (can be fractional)	2.00
monthly_salary	DECIMAL(19,4)	Y		Base monthly salary	4500.0000
variable_pay_pct	DECIMAL(5,4)	N		Variable pay as % of base	0.0500
annual_increment_pct	DECIMAL(5,4)	N		Annual pay increment	0.0500
benefits_pct	DECIMAL(5,4)	N		Benefits cost as % of salary	0.0800
hiring_trigger	VARCHAR(100)	N		Trigger condition for hiring	orders > 150/day
notes	TEXT	N		Rationale	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

headcount_plans

Scope: Core v1 | Grain: One headcount forecast per kitchen / role / period / scenario

Tracks planned headcount at the kitchen and role level over time. Derived from labor models and hiring triggers.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Assumption set	...
period_id	UUID	Y	planning_periods.id	Period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
kitchen_id	UUID	N	kitchens.id	Kitchen	...
role_name	VARCHAR(100)	Y		Role title	Delivery Rider
planned_headcount	DECIMAL(5,2)	Y		Planned FTEs	3.00
actual_headcount	DECIMAL(5,2)	N		Actual FTEs (if actuals loaded)	2.50
monthly_payroll	DECIMAL(19,4)	N		Computed total payroll for role/period	13500.0000
hiring_phase	VARCHAR(30)	N		LAUNCH|RAMP|STEADY_STATE	RAMP
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

opex_plans

Scope: Core v1 | Grain: One operating expense line per kitchen / period / scenario

Captures all non-COGS, non-payroll operating expenses including rent, utilities, tech SaaS, insurance, maintenance, and other fixed/variable overheads.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Assumption set	...
period_id	UUID	Y	planning_periods.id	Period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = corp)	...
category	VARCHAR(50)	Y		RENT|UTILITIES|TECH|INSURANCE|MAINTENANCE|OTHER	RENT
description	VARCHAR(200)	N		Line item description	Monthly kitchen rent
is_fixed	BOOLEAN	Y		Fixed vs variable cost	true
monthly_amount	DECIMAL(19,4)	Y		Expected monthly cost	11000.0000
escalation_rate	DECIMAL(5,4)	N		Annual escalation rate	0.0300
notes	TEXT	N		Notes	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

marketing_plans

Scope: Core v1 | Grain: One marketing budget line per channel / period / scenario

Captures planned marketing spend across channels including aggregator ads, social media, CRM, promotions, and influencer budgets.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Assumption set	...
period_id	UUID	Y	planning_periods.id	Period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
channel	VARCHAR(50)	Y		AGGREGATOR_ADS|SOCIAL|CRM|PROMO|INFLUENCER|OTHER	AGGREGATOR_ADS
budget	DECIMAL(19,4)	Y		Planned spend for period	8000.0000
expected_cac	DECIMAL(19,4)	N		Expected customer acquisition cost	25.0000
expected_uplift_pct	DECIMAL(5,4)	N		Demand uplift from this spend	0.0800
roi_target	DECIMAL(5,4)	N		Target marketing ROI	3.5000

notes	TEXT	N		Campaign notes	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

capex_plans

Scope: Core v1 | Grain: One capital expenditure item per kitchen / period / scenario

Schedules all capital expenditure including fit-out, equipment, technology, and vehicles. Feeds depreciation calculations and cash flow planning.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Assumption set	...
period_id	UUID	Y	planning_periods.id	Period of spend	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = corporate)	...
category	VARCHAR(50)	Y		FITOUT|EQUIPMENT|TECH|VEHICLE|OTHER	EQUIPMENT
description	VARCHAR(200)	N		Asset description	Pizza oven x2
amount	DECIMAL(19,4)	Y		Total capex amount	35000.0000
useful_life_months	INT	N		Asset useful life for depreciation	60
depreciation_method	VARCHAR(20)	N		STRAIGHT_LINE|DECLINING	STRAIGHT_LINE
monthly_depreciation	DECIMAL(19,4)	N		Computed monthly depreciation	583.3333
notes	TEXT	N		Procurement notes	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

working_capital_policies

Scope: Recommended v1.5 | Grain: One working capital policy per kitchen / scenario / period

Defines cash flow timing assumptions including inventory days, accounts receivable (platform payout days), and accounts payable terms. Drives the balance sheet and cash flow model.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
assumption_set_id	UUID	Y	assumption_sets.id	Assumption set	...
period_id	UUID	Y	planning_periods.id	Period	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = default)	...
inventory_days	INT	Y		Days of raw material inventory	3
accounts_receivable_days	INT	Y		Platform payout days (avg)	14
accounts_payable_days	INT	Y		Supplier payment terms	30
min_cash_reserve	DECIMAL(19,4)	N		Minimum operating cash balance	50000.0000
prepaid_months	INT	N		Months of rent/deposits prepaid	3
notes	TEXT	N		Policy notes	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

9. Layer E: Funding & Capital Strategy

equity_rounds

Scope: Core v1 | Grain: One fundraising event (seed, Series A, etc.)

Records equity investment events including round type, amount raised, pre-money valuation, and investor details. Feeds the cash flow and balance sheet models.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
round_name	VARCHAR(100)	Y		Round label	Series A
round_type	VARCHAR(30)	Y		SEED| PRE_SERIES_A| SERIES_A| SERIES_B| BRIDGE	SEED
target_amount	DECIMAL(19,4)	Y		Fundraising target	5000000.0000
amount_raised	DECIMAL(19,4)	N		Actual amount raised	5000000.0000
pre_money_valuation	DECIMAL(19,4)	N		Pre-money valuation	20000000.0000
post_money_valuation	DECIMAL(19,4)	N		Post-money valuation	25000000.0000
close_date	DATE	N		Expected or actual close date	2026-06-01
dilution_pct	DECIMAL(5,4)	N		Ownership % sold in this round	0.2000
use_of_proceeds	JSONB	N		Planned allocation of funds	{"expansion":0.6,"opex":0.4}
notes	TEXT	N		Investor notes	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

debt_facilities

Scope: Core v1 | Grain: One debt instrument / loan facility

Captures debt financing instruments including bank loans, revolving credit, and convertible notes. Used to model interest expense and debt service.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
facility_name	VARCHAR(100)	Y		Facility label	Emirates NBD Term Loan
type	VARCHAR(30)	Y		TERM_LOAN| REVOLVING| CONVERTIBLE| OVERDRAFT	TERM_LOAN
principal_amount	DECIMAL(19,4)	Y		Total facility amount	2000000.0000
drawn_amount	DECIMAL(19,4)	N		Amount drawn at start	2000000.0000

interest_rate	DECIMAL(5,4)	Y		Annual interest rate	0.0650
rate_type	VARCHAR(20)	Y		FIXED|VARIABLE	FIXED
start_date	DATE	Y		Drawdown/ facility start date	2026-01-01
maturity_date	DATE	Y		Repayment maturity date	2029-01-01
covenant_notes	TEXT	N		Key covenant conditions	DSCR min 1.2x
notes	TEXT	N		Additional terms	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

debt_repayment_schedules

Scope: Core v1 | Grain: One amortization entry per period per debt facility

Stores the amortization schedule for each debt facility on a period-by-period basis. Used to compute principal repayments, interest, and outstanding balance.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
debt_facility_id	UUID	Y	debt_facilities.id	Parent facility	...
period_id	UUID	Y	planning_periods .id	Period	...
opening_balance	DECIMAL(19,4)	Y		Outstanding principal start	2000000.0000
principal_repayment	DECIMAL(19,4)	Y		Principal paid this period	55555.5556
interest_charge	DECIMAL(19,4)	Y		Interest for period	10833.3333
closing_balance	DECIMAL(19,4)	Y		Outstanding principal end	1944444.4444
is_balloon_payment	BOOLEAN	N		True if final balloon payment	false
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

capital_allocation_plans

Scope: Phase 2-3 | Grain: One capital deployment rule per market / period / scenario

Defines how capital (equity and debt) is allocated across markets and kitchens over time. Enables portfolio-level capital optimization.

Field Name	Type	Req	FK / Reference	Description	Example

id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	Y	planning_periods.id	Period	...
market_scope_type	VARCHAR(20)	Y		MICRO|MACRO|CLUSTER|COUNTRY	MICRO
market_scope_id	UUID	Y		FK to relevant geo entity	...
capex_allocation	DECIMAL(19,4)	N		Capital allocated for CAPEX	350000.0000
opex_runway_months	INT	N		Operating runway funded	12
priority_score	DECIMAL(5,2)	N		Allocation priority rank	85.00
notes	TEXT	N		Rationale	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

dilution_models

Scope: Recommended v1.5 | Grain: One cap table impact model per equity round / scenario

Models ownership dilution resulting from equity rounds, option pools, and convertible instruments. Supports founder and investor equity planning.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
equity_round_id	UUID	N	equity_rounds.id	Linked equity round	...
period_id	UUID	Y	planning_periods.id	Period	...
pre_round_shares	BIGINT	N		Total shares before round	10000000
new_shares_issued	BIGINT	N		Shares issued in round	2500000
option_pool_shares	BIGINT	N		ESOP pool shares	500000
founder_ownership_pct	DECIMAL(5,4)	N		Founder % post-round	0.5000
investor_ownership_pct	DECIMAL(5,4)	N		New investor % post-round	0.2000
implied_valuation	DECIMAL(19,4)	N		Implied company valuation	25000000.0000
notes	TEXT	N		Commentary	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

10. Layer F: Financial Projection Engine

pnl_projections

Scope: Core v1 | Grain: One P&L line per period / kitchen / scenario

The central financial output table. Stores the full projected Income Statement from gross revenue through to net income and EBITDA for each period and market scope.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	Y	planning_periods.id	Period	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = rolled up)	...
gross_revenue	DECIMAL(19,4)	Y		Total sales before discounts	120000.0000
discounts	DECIMAL(19,4)	Y		Discount value	12000.0000
platform_commissions	DECIMAL(19,4)	Y		Aggregator fees	24000.0000
net_revenue	DECIMAL(19,4)	Y		Gross - Discounts - Commissions	84000.0000
cogs_food	DECIMAL(19,4)	Y		Food ingredient cost	21000.0000
cogs_packaging	DECIMAL(19,4)	Y		Packaging cost	2800.0000
cogs_other	DECIMAL(19,4)	N		Other variable COGS	1400.0000
gross_profit	DECIMAL(19,4)	Y		Net Revenue - Total COGS	58800.0000
payroll	DECIMAL(19,4)	Y		Total staff cost	18000.0000
rent	DECIMAL(19,4)	Y		Rent expense	11000.0000
utilities	DECIMAL(19,4)	N		Utility expense	2500.0000
marketing_spend	DECIMAL(19,4)	N		Total marketing cost	5000.0000
tech_saas	DECIMAL(19,4)	N		Technology and SaaS subscriptions	1200.0000
other_opex	DECIMAL(19,4)	N		Other operating expenses	2000.0000
total_opex	DECIMAL(19,4)	Y		Sum of all operating expenses	39700.0000
ebitda	DECIMAL(19,4)	Y		Gross Profit - Total OPEX	19100.0000
depreciation	DECIMAL(19,4)	N		D&A charge	2500.0000
ebit	DECIMAL(19,4)	N		EBITDA - Depreciation	16600.0000
interest_expense	DECIMAL(19,4)	N		Debt interest charge	1083.3333
tax_charge	DECIMAL(19,4)	N		Corporate tax	1394.1000
net_income	DECIMAL(19,4)	Y		EBIT - Interest - Tax	14122.5667
ebitda_margin	DECIMAL(5,4)	N		EBITDA / Net Revenue	0.2274
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

cashflow_projections

Scope: Core v1 | Grain: One cash flow statement per period / scenario

Projects cash inflows and outflows from operations, investing (capex), and financing (funding/debt). Shows opening balance, net movement, and closing cash position.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	Y	planning_periods.id	Period	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = company level)	...
opening_cash	DECIMAL(19,4)	Y		Cash at start of period	200000.0000
operating_inflows	DECIMAL(19,4)	Y		Cash collected from customers	84000.0000
cogs_paid	DECIMAL(19,4)	Y		Cash paid for ingredients	25200.0000
payroll_paid	DECIMAL(19,4)	Y		Cash paid to staff	18000.0000
opex_paid	DECIMAL(19,4)	Y		Other operating cash outflows	21700.0000
net_operating_cf	DECIMAL(19,4)	Y		Operating inflows - outflows	19100.0000
capex_outflows	DECIMAL(19,4)	N		Capital expenditures	5000.0000
net_investing_cf	DECIMAL(19,4)	N		Net investing cash flow	-5000.0000
equity_raised	DECIMAL(19,4)	N		Proceeds from equity	0.0000
debt_drawn	DECIMAL(19,4)	N		New debt drawdowns	0.0000
debt_repaid	DECIMAL(19,4)	N		Principal repayments	55555.5556
interest_paid	DECIMAL(19,4)	N		Interest payments	10833.3333
net_financing_cf	DECIMAL(19,4)	N		Net financing cash flow	-66388.8889
net_cash_movement	DECIMAL(19,4)	Y		Total net cash change	-52288.8889
closing_cash	DECIMAL(19,4)	Y		Cash at end of period	147711.1111
cash_runway_months	DECIMAL(5,2)	N		Months of runway at current burn	11.50
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

balance_sheet_projections

Scope: Core v1 | Grain: One balance sheet snapshot per period / scenario

Projects the balance sheet position at each period end, including assets, liabilities, and equity. Validates with the accounting equation A = L + E.

Field Name	Type	Req	FK / Reference	Description	Example

unit_economics_projections

Scope: Core v1 | Grain: One per-order economics record per period / kitchen / scenario

Computes all key per-order metrics: AOV, contribution margin, EBITDA per order, payback period. Enables bottom-up unit economics validation.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	Y	planning_periods.id	Period	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = portfolio avg)	...
orders	DECIMAL(19,4)	Y		Total orders in period	1732.5000
avg_order_value	DECIMAL(19,4)	Y		Gross AOV	52.5000
net_revenue_per_order	DECIMAL(19,4)	Y		Net revenue per order	48.50
cogs_per_order	DECIMAL(19,4)	Y		COGS per order	13.00
contribution_per_order	DECIMAL(19,4)	Y		Net rev - COGS per order	35.50
fixed_cost_per_order	DECIMAL(19,4)	N		Allocated fixed cost per order	22.90
ebitda_per_order	DECIMAL(19,4)	Y		EBITDA / total orders	11.02
contribution_margin_pct	DECIMAL(5,4)	N		Contribution / Net Rev	0.7320
ebitda_margin_pct	DECIMAL(5,4)	N		EBITDA margin	0.2274
payback_months	DECIMAL(5,2)	N		Months to recover capex	18.50
cac	DECIMAL(19,4)	N		Customer acquisition cost	25.0000
clv_12m	DECIMAL(19,4)	N		12-month customer lifetime value	320.0000
clv_cac_ratio	DECIMAL(5,4)	N		CLV / CAC	12.8000
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

11. Layer G: KPI & Performance Intelligence

kpi_projections

Scope: Core v1 | Grain: One KPI record per metric / period / scenario

Stores all calculated KPI values including financial ratios (ROIC, IRR, NPV), operational metrics, and performance flags. Used in executive dashboards.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	N	planning_periods .id	Period (null = lifetime metric)	...
kitchen_id	UUID	N	kitchens.id	Kitchen (null = portfolio)	...
kpi_code	VARCHAR(50)	Y		Unique KPI identifier	EBITDA_MARGIN
kpi_name	VARCHAR(100)	Y		Human-readable name	EBITDA Margin %
value	DECIMAL(19,6)	Y		Calculated KPI value	0.227400
unit	VARCHAR(20)	N		PCT|AED|RATIO| COUNT|MONTHS	PCT
target_value	DECIMAL(19,6)	N		Target benchmark	0.200000
is_on_target	BOOLEAN	N		True if value >= target	true
variance_pct	DECIMAL(5,4)	N		Variance from target	0.0137
alert_threshold	DECIMAL(19,6)	N		Below this = red alert	0.100000
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

driver_explainability

Scope: Recommended v1.5 | Grain: One explainability record per KPI / driver pair / period

Attributes KPI movements to specific drivers using contribution analysis. Enables 'why did EBITDA change?' explanations in dashboards.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	Y	planning_periods .id	Period	...
kpi_code	VARCHAR(50)	Y		KPI being explained	EBITDA
driver_name	VARCHAR(100)	Y		Driver causing change	growth_rate
driver_table	VARCHAR(100)	N		Source table of driver	demand_drivers
contribution_amount	DECIMAL(19,4)	Y		AED contribution of this driver	5000.0000
contribution_pct	DECIMAL(5,4)	Y		% share of total KPI movement	0.2600

direction	VARCHAR(10)	Y		POSITIVE| NEGATIVE| NEUTRAL	POSITIVE
notes	TEXT	N		Explanation narrative	Growth rate increase drove higher orders
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

variance_attribution

Scope: Recommended v1.5 | Grain: One variance bridge entry per period / scenario pair

Stores the variance bridge between two scenarios (e.g. Budget vs Forecast) or between plan and actuals. Used in period-end management reporting.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
base_scenario_id	UUID	Y	scenarios.id	Reference scenario	...
compare_scenario_id	UUID	Y	scenarios.id	Comparison scenario	...
period_id	UUID	Y	planning_periods.id	Period	...
kpi_code	VARCHAR(50)	Y		KPI being compared	EBITDA
base_value	DECIMAL(19,4)	Y		Value in base scenario	19100.0000
compare_value	DECIMAL(19,4)	Y		Value in comparison scenario	22500.0000
variance_amount	DECIMAL(19,4)	Y		Compare - Base	3400.0000
variance_pct	DECIMAL(5,4)	Y		% variance	0.1780
primary_driver	VARCHAR(100)	N		Main attributable driver	Higher orders
notes	TEXT	N		Bridge narrative	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

performance_alerts

Scope: Recommended v1.5 | Grain: One alert event per KPI breach

Triggered when a KPI breaches a defined threshold. Feeds the Trigger & Alert dashboard and supports automated notification workflows.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	Y	planning_periods .id	Period of breach	...
kpi_code	VARCHAR(50)	Y		KPI that triggered alert	CASH_RUNWAY
alert_type	VARCHAR(30)	Y		BELOW_THRESH OLD| ABOVE_THRESH OLD|FLAT	BELOW_THRESH OLD
severity	VARCHAR(20)	Y		INFO|WARNING| CRITICAL	CRITICAL
threshold_value	DECIMAL(19,6)	Y		Threshold that was breached	3.000000
actual_value	DECIMAL(19,6)	Y		Actual KPI value at breach	2.500000
message	TEXT	N		Alert message	Cash runway below 3 months
resolved_at	TIMESTAMPTZ	N		Timestamp when resolved	
resolved_by	UUID	N	users.id	Resolver user	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

12. Layer H: Risk & Uncertainty Framework

risk_objects

Scope: Recommended v1.5 | Grain: One defined risk factor

Catalog of identified risk factors that could affect projections. Each risk object defines the risk type, affected KPIs, probability, and impact range.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
name	VARCHAR(100)	Y		Risk name	Platform commission increase
category	VARCHAR(50)	Y		MARKET| REGULATORY| OPERATIONAL| FINANCIAL| COMPETITIVE	MARKET
description	TEXT	N		Full risk description	Aggregators raise commission by 2-5%
affected_kpis	JSONB	N		List of KPIs affected	["EBITDA_MARGIN","NET_REVENUE"]
probability	DECIMAL(5,4)	N		Likelihood 0.00-1.00	0.4000
impact_low	DECIMAL(19,4)	N		Downside impact (AED)	-50000.0000
impact_high	DECIMAL(19,4)	N		Upside impact if applicable	0.0000
severity	VARCHAR(20)	N		LOW|MEDIUM| HIGH|CRITICAL	HIGH
mitigation_strategy	TEXT	N		How to mitigate	Develop direct ordering channel
owner_id	UUID	N	users.id	Risk owner	...
review_date	DATE	N		Next review date	2026-04-01
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

risk_scenarios

Scope: Recommended v1.5 | Grain: One risk-adjusted scenario modification record

Links risk objects to scenario modifications. Defines how a risk event changes driver assumptions, enabling scenario-based risk impact analysis.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Parent scenario	...
risk_object_id	UUID	Y	risk_objects.id	Linked risk	...
period_id	UUID	N	planning_periods.id	Period of risk impact	...
driver_table	VARCHAR(100)	Y		Table with assumption to modify	demand_drivers

driver_field	VARCHAR(100)	Y		Field to modify	growth_rate
adjustment_type	VARCHAR(20)	Y		ABSOLUTE| PERCENTAGE| REPLACE	PERCENTAGE
adjustment_value	DECIMAL(10,6)	Y		Magnitude of adjustment	-0.030000
is_active	BOOLEAN	Y		Whether this adjustment is live	true
notes	TEXT	N		Risk scenario narrative	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

monte_carlo_runs

Scope: Phase 2-3 | Grain: One Monte Carlo simulation execution

Stores metadata for each Monte Carlo simulation run including number of iterations, random seed, and targeted KPIs.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Base scenario for simulation	...
run_name	VARCHAR(100)	Y		Run label	MC Run Jan 2026
iterations	INT	Y		Number of simulation iterations	10000
random_seed	INT	N		Seed for reproducibility	42
target_kpis	JSONB	Y		KPIs to simulate	["EBITDA","IRR"]
run_status	VARCHAR(20)	Y		QUEUED| RUNNING| COMPLETED| FAILED	COMPLETED
started_at	TIMESTAMPTZ	N		Run start timestamp	2026-01-15 08:00+04
completed_at	TIMESTAMPTZ	N		Run end timestamp	2026-01-15 08:05+04
notes	TEXT	N		Run notes	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

monte_carlo_summaries

Scope: Phase 2-3 | Grain: One statistical summary per KPI per MC run

Aggregated probabilistic results from Monte Carlo runs. Stores percentile outcomes, expected values, and tail-risk metrics for each KPI.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
run_id	UUID	Y	monte_carlo_runs.id	Parent MC run	...
kpi_code	VARCHAR(50)	Y		KPI analyzed	EBITDA
mean_value	DECIMAL(19,4)	Y		Mean outcome	19100.0000
std_dev	DECIMAL(19,4)	Y		Standard deviation	4500.0000
p10_value	DECIMAL(19,4)	N		10th percentile outcome	12000.0000
p25_value	DECIMAL(19,4)	N		25th percentile outcome	15500.0000
p50_value	DECIMAL(19,4)	N		Median outcome	19000.0000
p75_value	DECIMAL(19,4)	N		75th percentile outcome	22500.0000
p90_value	DECIMAL(19,4)	N		90th percentile outcome	26000.0000
prob_positive	DECIMAL(5,4)	N		Probability of positive outcome	0.7800
var_95	DECIMAL(19,4)	N		Value at Risk 95% confidence	-8000.0000
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

trigger_log

Scope: Recommended v1.5 | Grain: One trigger event fired by the rules engine

Records every trigger event fired by the planning engine (e.g. launch a new kitchen, hire additional staff, raise capital). Provides an audit trail of automated decisions.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario context	...
period_id	UUID	Y	planning_periods.id	Period when triggered	...
trigger_type	VARCHAR(50)	Y		KITCHEN_LAUNCH|HIRING|CAPEX|FUNDING|ALERT	HIRING
trigger_rule	VARCHAR(200)	N		Rule condition that fired	daily_orders > 150
entity_affected	VARCHAR(100)	N		Affected entity type and ID	kitchen:DXB-JLT-01
action_taken	TEXT	N		Description of resulting action	Hired 1 additional chef
impact_amount	DECIMAL(19,4)	N		Financial impact of action	4500.0000
fired_at	TIMESTAMPTZ	Y		Timestamp when trigger fired	2026-02-01 00:00+04
reviewed_by	UUID	N	users.id	User who reviewed	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...

created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

13. Layer I: Simulation & Decision Support

simulation_scenarios

Scope: Phase 2-3 | Grain: One temporary what-if simulation context

Stores temporary what-if simulation contexts created in the Simulation Lab. Unlike formal scenarios, these are transient and can be discarded after analysis.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
parent_scenario_id	UUID	Y	scenarios.id	Base scenario being modified	...
name	VARCHAR(100)	Y		Simulation name	Price +10% Test
description	TEXT	N		What is being simulated	Impact of 10% price increase
is_transient	BOOLEAN	Y		True = temporary, delete on expiry	true
expires_at	TIMESTAMPTZ	N		Auto-delete timestamp	2026-01-16 23:59+04
created_by_session	VARCHAR(100)	N		Session or user context	session_abc123
overrides	JSONB	N		Driver value overrides applied	{"list_price": 1.10}
notes	TEXT	N		Notes	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

simulation_results

Scope: Phase 2-3 | Grain: One transient KPI output per simulation scenario / period

Stores output KPIs computed from simulation scenarios. Transient records used to populate the Simulation Lab output panels.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
simulation_id	UUID	Y	simulation_scenarios.id	Parent simulation	...
period_id	UUID	N	planning_periods.id	Period (null = lifetime)	...
kpi_code	VARCHAR(50)	Y		KPI calculated	EBITDA
value	DECIMAL(19,6)	Y		Simulated KPI value	23500.000000
base_value	DECIMAL(19,6)	N		Base scenario KPI for comparison	19100.000000
delta	DECIMAL(19,6)	N		Simulation - Base	4400.000000
delta_pct	DECIMAL(5,4)	N		% change	0.2304
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04

updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

sensitivity_analyses

Scope: Phase 2-3 | Grain: One sensitivity output per driver variable / KPI pair

Stores results of one-at-a-time sensitivity analysis showing how much a KPI changes when a specific driver is varied across a defined range.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario context	...
kpi_code	VARCHAR(50)	Y		Target KPI	EBITDA
driver_name	VARCHAR(100)	Y		Driver being varied	growth_rate
driver_table	VARCHAR(100)	N		Source table of driver	demand_drivers
driver_field	VARCHAR(100)	N		Field being varied	growth_rate
test_range_low	DECIMAL(10,6)	Y		Lower bound of test range	-0.100000
test_range_high	DECIMAL(10,6)	Y		Upper bound of test range	0.150000
base_kpi_value	DECIMAL(19,4)	Y		KPI at base assumption	19100.0000
low_kpi_value	DECIMAL(19,4)	Y		KPI at low range	11000.0000
high_kpi_value	DECIMAL(19,4)	Y		KPI at high range	28500.0000
elasticity	DECIMAL(10,6)	N		% KPI change per % driver change	2.500000
tornado_rank	INT	N		Rank in tornado chart (1 = highest)	1
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

decision_memory

Scope: Phase 2-3 | Grain: One historical decision record

Archives significant planning decisions made during scenario modeling. Enables institutional memory, pattern recognition, and future decision support.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	N	scenarios.id	Associated scenario	...

decision_type	VARCHAR(50)	Y		EXPANSION|PRICING|HIRING|FUNDING|OTHER	EXPANSION
decision_title	VARCHAR(200)	Y		Brief decision description	Launch JLT Cluster A kitchen
decision_rationale	TEXT	N		Why the decision was made	Attractiveness score 74.5 > threshold
alternatives_considered	TEXT	N		Other options evaluated	Considered JLT Cluster B
expected_impact	JSONB	N		Projected financial impact	{"ebitda_uplift": 50000}
decided_by	UUID	N	users.id	Decision maker	...
decided_at	TIMESTAMPTZ	Y		When decision was made	2026-01-10 14:00+04
actual_outcome	TEXT	N		Outcome after the fact	Kitchen opened 2 months late
outcome_recorded_at	TIMESTAMPTZ	N		When outcome was recorded	2026-04-01
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

decision_log

Scope: Phase 2-3 | Grain: One audit entry per user action in the planning UI

Detailed audit trail of user actions within the planning engine: assumption edits, scenario changes, approvals, simulations run. Supports compliance and change management.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
user_id	UUID	Y	users.id	User performing action	...
scenario_id	UUID	N	scenarios.id	Scenario context	...
action_type	VARCHAR(50)	Y		EDIT|CREATE|DELETE|APPROVE|RUN_SIM	EDIT
entity_type	VARCHAR(50)	Y		Entity type modified	demand_drivers
entity_id	UUID	N		Entity ID	...
description	TEXT	N		What changed	growth_rate changed 0.05->0.07
ip_address	INET	N		User IP for security audit	10.0.0.1
session_id	VARCHAR(100)	N		Session identifier	sess_abc123
occurred_at	TIMESTAMPTZ	Y		Action timestamp	2026-01-12 11:00+04
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...

deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

14. Layer J: Portfolio Construction

portfolio_plans

Scope: Phase 2-3 | Grain: One multi-market portfolio strategy per scenario

Defines a portfolio-level expansion strategy covering multiple markets. Used for prioritizing market entry, capital allocation, and portfolio-level IRR/NPV optimization.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
name	VARCHAR(100)	Y		Portfolio plan name	UAE 2026-2028 Expansion
target_kitchens	INT	N		Target number of kitchens	12
target_markets	INT	N		Target number of micros	12
capital_budget	DECIMAL(19,4)	N		Total capital available for expansion	8000000.0000
target_irr	DECIMAL(5,4)	N		Target portfolio IRR	0.3500
target_payback_months	INT	N		Target payback period	24
strategy	TEXT	N		Portfolio strategy narrative	Hub-and-spoke expansion...
status	VARCHAR(20)	N		DRAFT|ACTIVE|APPROVED|COMPLETED	ACTIVE
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

market_portfolio_scores

Scope: Phase 2-3 | Grain: One scoring record per micro area per portfolio plan

Stores attractiveness and performance scores for each market in a portfolio plan. Drives market prioritization and launch sequencing.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
portfolio_plan_id	UUID	Y	portfolio_plans.id	Parent portfolio plan	...
micro_id	UUID	Y	micros.id	Micro area being scored	...
attractiveness_score	DECIMAL(5,2)	N		Composite attractiveness 0-100	74.50
readiness_score	DECIMAL(5,2)	N		Operational readiness 0-100	68.00
expected_irr	DECIMAL(5,4)	N		Projected IRR for this market	0.3200
expected_payback_months	INT	N		Projected payback in months	20

priority_rank	INT	N		Launch priority rank	3
recommended_launch_period_id	UUID	N	planning_periods.id	Recommended launch period	...
notes	TEXT	N		Scoring commentary	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

portfolio_capital_allocations

Scope: Phase 2-3 | Grain: One capital allocation per market per portfolio plan per period

Distributes the portfolio capital budget across individual markets and time periods. Enables capital-constrained expansion modeling.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
portfolio_plan_id	UUID	Y	portfolio_plans.id	Parent portfolio plan	...
micro_id	UUID	Y	micros.id	Target market	...
period_id	UUID	Y	planning_periods.id	Allocation period	...
capex_allocation	DECIMAL(19,4)	N		CAPEX allocated	350000.0000
opex_allocation	DECIMAL(19,4)	N		Working capital / opex support	120000.0000
total_allocation	DECIMAL(19,4)	N		Total capital for this market/period	470000.0000
allocation_type	VARCHAR(30)	N		LAUNCH| EXPANSION| MAINTENANCE	LAUNCH
notes	TEXT	N		Allocation rationale	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

portfolio_return_metrics

Scope: Phase 2-3 | Grain: One aggregated return metric per portfolio plan

Stores portfolio-level financial return metrics including aggregate IRR, NPV, total capital deployed, and MOIC. Used in the Portfolio Optimization console.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...

portfolio_plan_id	UUID	Y	portfolio_plans.id	Parent portfolio plan	...
scenario_id	UUID	Y	scenarios.id	Scenario	...
period_id	UUID	N	planning_periods.id	Period (null = lifetime)	...
total_capital_deployed	DECIMAL(19,4)	N		Total capital invested	8000000.0000
portfolio_npv	DECIMAL(19,4)	N		Aggregate NPV at hurdle rate	12500000.0000
portfolio_irr	DECIMAL(5,4)	N		Blended portfolio IRR	0.3400
portfolio_moic	DECIMAL(5,4)	N		Multiple on invested capital	2.5600
avg_payback_months	DECIMAL(5,2)	N		Average payback across portfolio	21.50
active_kitchens	INT	N		Number of active kitchens	8
total_revenue_annualized	DECIMAL(19,4)	N		Annualized portfolio revenue	15000000.0000
total_ebitda_annualized	DECIMAL(19,4)	N		Annualized portfolio EBITDA	3200000.0000
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

15. Optional Extension Layer

These 11 tables capture operational detail not required for strategic financial planning. They are deferred to Phase 6 (Week 61+) and activated only when operational intelligence is needed for actuals reconciliation, scheduling optimization, or recipe-level costing.

order_headers

Scope: Optional Extension | Grain: One raw transaction / delivery order

Raw order transaction header. Deferred to Phase 6 / Actuals Bridge. Not required for strategic planning but enables operational drill-down and actuals reconciliation.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
kitchen_id	UUID	Y	kitchens.id	Originating kitchen	...
platform_id	UUID	Y	platforms.id	Sales channel	...
order_placed_at	TIMESTAMPTZ	Y		Order timestamp	2026-01-15 19:30+04
gross_amount	DECIMAL(19,4)	Y		Gross order value	55.0000
discount_amount	DECIMAL(19,4)	N		Discount applied	5.0000
net_amount	DECIMAL(19,4)	Y		Net amount	50.0000
status	VARCHAR(20)	Y		PLACED| CONFIRMED| DELIVERED| CANCELLED	DELIVERED
delivery_time_min	INT	N		Delivery time in minutes	28
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

order_lines

Scope: Optional Extension | Grain: One line item within a raw order

Individual product lines within an order header. Enables product-level actuals analysis and SKU-level performance tracking in Phase 6.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
order_id	UUID	Y	order_headers.id	Parent order	...
product_family_id	UUID	N	product_families.id	Product family	...
product_name	VARCHAR(100)	Y		Product name at time	Pepperoni Pizza
quantity	INT	Y		Units ordered	1
unit_price	DECIMAL(19,4)	Y		Price per unit	45.0000
line_total	DECIMAL(19,4)	Y		Qty * unit price	45.0000
food_cost	DECIMAL(19,4)	N		Actual food cost	11.2500
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04

updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

kitchen_stations

Scope: Optional Extension | Grain: One production station within a kitchen

Models kitchen production stations (prep, oven, packing). Used in Phase 6 for operational capacity analysis and bottleneck identification.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
kitchen_id	UUID	Y	kitchens.id	Parent kitchen	...
name	VARCHAR(100)	Y		Station name	Pizza Oven 1
type	VARCHAR(30)	Y		PREP|OVEN| PACKING| DISPATCH	OVEN
capacity_units_per_hr	INT	N		Max units per hour	24
is_bottleneck	BOOLEAN	N		Identified bottleneck	false
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

staff_roster

Scope: Optional Extension | Grain: One actual staff shift entry

Records actual staff shift data at the kitchen level. Used in Phase 6 for actuals vs planned headcount reconciliation.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
kitchen_id	UUID	Y	kitchens.id	Kitchen	...
role_name	VARCHAR(100)	Y		Role	Pizza Chef
shift_date	DATE	Y		Shift date	2026-01-15
hours_worked	DECIMAL(4,2)	N		Actual hours worked	8.00
actual_cost	DECIMAL(19,4)	N		Actual shift cost	150.0000
status	VARCHAR(20)	Y		WORKED| ABSENT| OVERTIME	WORKED
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04

created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

actual_ingestion_log

Scope: Optional Extension | Grain: One data import event

Tracks when and how actuals data was ingested from external systems (POS, aggregator portals). Used for data lineage and reconciliation.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
source_system	VARCHAR(50)	Y		POS| AGGREGATOR| BANK|MANUAL	AGGREGATOR
source_name	VARCHAR(100)	N		System name	Talabat Portal
import_type	VARCHAR(50)	Y		ORDERS| REVENUE| PAYROLL|CAPEX	ORDERS
records_imported	INT	Y		Count of records loaded	1842
period_start	DATE	Y		Data coverage start	2026-01-01
period_end	DATE	Y		Data coverage end	2026-01-31
status	VARCHAR(20)	Y		SUCCESS| PARTIAL|FAILED	SUCCESS
error_notes	TEXT	N		Error details if partial/failed	
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

intraday_demand_buckets

Scope: Optional Extension | Grain: One 30-minute or hourly demand bucket

Captures intraday demand patterns at 30-minute or hourly granularity. Used in Phase 6 for scheduling optimization and peak demand analysis.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
kitchen_id	UUID	Y	kitchens.id	Kitchen	...
date	DATE	Y		Date of bucket	2026-01-15
bucket_start_time	TIME	Y		Bucket start time	19:00:00
bucket_end_time	TIME	Y		Bucket end time	19:30:00
expected_orders	INT	N		Forecast orders in bucket	12
actual_orders	INT	N		Actual orders in bucket	14
day_of_week	INT	Y		1=Mon to 7=Sun	4

is_peak	BOOLEAN	N		Identified peak bucket	true
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

operational_events

Scope: Optional Extension | Grain: One operational incident or event at a kitchen

Logs operational events including outages, equipment failures, and supply disruptions. Used in Phase 6 for operational reliability analysis.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
kitchen_id	UUID	Y	kitchens.id	Affected kitchen	...
event_type	VARCHAR(50)	Y		OUTAGE| EQUIPMENT| SUPPLY| STAFFING| OTHER	EQUIPMENT
description	TEXT	N		Event description	Oven 1 breakdown
occurred_at	TIMESTAMPTZ	Y		Event timestamp	2026-01-20 11:30+04
resolved_at	TIMESTAMPTZ	N		Resolution timestamp	2026-01-20 14:00+04
revenue_impact	DECIMAL(19,4)	N		Estimated revenue loss	3500.0000
severity	VARCHAR(20)	Y		LOW|MEDIUM| HIGH|CRITICAL	HIGH
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

product_variants

Scope: Optional Extension | Grain: One SKU variant within a product family

Captures individual SKU variations (sizes, crust types) within a product family. Deferred to Phase 6; not required for strategic financial planning.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
product_family_id	UUID	Y	product_families.id	Parent family	...
name	VARCHAR(100)	Y		Variant name	Pepperoni Large
sku	VARCHAR(50)	Y		SKU code	PIZ-PEP-LG

size	VARCHAR(30)	N		Size/format	Large
list_price	DECIMAL(19,4)	Y		Listed price	52.0000
food_cost	DECIMAL(19,4)	N		Actual food cost	13.0000
prep_time_min	INT	N		Preparation time minutes	12
is_active	BOOLEAN	Y		Active on menu	true
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

modifier_types

Scope: Optional Extension | Grain: One customization/add-on option

Defines product modifiers (extra toppings, crust choices). Deferred to Phase 6; not required for driver-based financial projections.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
product_family_id	UUID	Y	product_families.id	Parent family	...
name	VARCHAR(100)	Y		Modifier name	Extra Cheese
price_add_on	DECIMAL(19,4)	N		Additional price	5.0000
cost_add_on	DECIMAL(19,4)	N		Additional food cost	1.5000
is_active	BOOLEAN	Y		Active flag	true
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

ingredient_bom

Scope: Optional Extension | Grain: One ingredient line in a product bill of materials

Bill of materials for product variants. Enables recipe-level costing and supplier cost management in Phase 6.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
product_variant_id	UUID	Y	product_variants.id	Parent variant	...
ingredient_name	VARCHAR(100)	Y		Ingredient name	Mozzarella
quantity	DECIMAL(10,4)	Y		Quantity used per unit	0.1500
unit	VARCHAR(20)	Y		KG|LITRE|PIECE|GRAM	KG
cost_per_unit	DECIMAL(19,4)	Y		Cost per unit of ingredient	45.0000

supplier	VARCHAR(100)	N		Supplier name	Al Ain Foods
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

recipe_histories

Scope: Optional Extension | Grain: One versioned recipe record per product variant

Tracks recipe versions over time enabling cost impact analysis when recipes change. Phase 6 feature for R&D and menu engineering teams.

Field Name	Type	Req	FK / Reference	Description	Example
id	UUID	Y		Primary key	...
product_variant_id	UUID	Y	product_variants.id	Product variant	...
version	VARCHAR(20)	Y		Recipe version label	v1.2
effective_from	DATE	Y		When this version became active	2026-01-01
effective_to	DATE	N		When superseded (null=current)	
total_food_cost	DECIMAL(19,4)	Y		Total food cost per unit	13.0000
change_notes	TEXT	N		What changed and why	Reduced cheese quantity
approved_by	UUID	N	users.id	Approval user	...
company_id	UUID	Y	companies.id	Tenant isolation key	550e8400...
created_at	TIMESTAMPTZ	Y		Record creation timestamp	2026-01-01 09:00:00+04
updated_at	TIMESTAMPTZ	Y		Last modification timestamp	2026-01-15 14:30:00+04
created_by	UUID	Y	users.id	User who created the record	a1b2c3d4...
updated_by	UUID	Y	users.id	User who last updated the record	a1b2c3d4...
deleted_at	TIMESTAMPTZ	N		Soft-delete timestamp; NULL = active	

16. Enum Reference

All enumerated string values used across the schema. Enforce at application layer; store as VARCHAR with check constraints or native ENUM types where supported.

Enum Code	Used In (Table.Field)	Allowed Values
SCENARIO_TYPE	scenarios.type	BASE, UPSIDE, DOWNSIDE, STRESS, BUDGET, FORECAST
ASSUMPTION_STATUS	assumption_sets.status	DRAFT, ACTIVE, ARCHIVED
CONFIDENCE_LEVEL	assumption_sets.confidence_level	HIGH, MEDIUM, LOW, SPECULATIVE
APPROVAL_STATUS	governance_events.event_type	APPROVAL, REJECTION, LOCK, UNLOCK, COMMENT
PERIOD_TYPE	planning_calendars.period_type	MONTHLY, QUARTERLY, WEEKLY
UNIT_STATUS	kitchens.status	PLANNED, CONSTRUCTION, ACTIVE, CLOSED, PAUSED
PLATFORM_TYPE	platforms.type	AGGREGATOR, DIRECT, B2B, DINE_IN
CATEGORY_TYPE	opex_plans.category	RENT, UTILITIES, TECH, INSURANCE, MAINTENANCE, OTHER
CAPEX_CATEGORY	capex_plans.category	FITOUT, EQUIPMENT, TECH, VEHICLE, OTHER
MARKETING_CHANNEL	marketing_plans.channel	AGGREGATOR_ADS, SOCIAL, CRM, PROMO, INFLUENCER, OTHER
RISK_CATEGORY	risk_objects.category	MARKET, REGULATORY, OPERATIONAL, FINANCIAL, COMPETITIVE
SEVERITY_LEVEL	risk_objects.severity	LOW, MEDIUM, HIGH, CRITICAL
TRIGGER_TYPE	trigger_log.trigger_type	KITCHEN_LAUNCH, HIRING, CAPEX, FUNDING, ALERT
DEBT_TYPE	debt_facilities.type	TERM_LOAN, REVOLVING, CONVERTIBLE, OVERDRAFT
MARKET_STATUS	micros.launch_status	PIPELINE, PLANNED, ACTIVE, PAUSED, CLOSED
SCOPE_TAXONOMY	various	CORE_V1, RECOMMENDED_V1_5, PHASE2_3, OPTIONAL_EXTENSION
ACTION_TYPE	decision_log.action_type	EDIT, CREATE, DELETE, APPROVE, RUN_SIM
MC_STATUS	monte_carlo_runs.run_status	QUEUED, RUNNING, COMPLETED, FAILED
ALLOCATION_TYPE	portfolio_capital_allocations.allocation_type	LAUNCH, EXPANSION, MAINTENANCE
DECISION_TYPE	decision_memory.decision_type	EXPANSION, PRICING, HIRING, FUNDING, OTHER
ALERT_SEVERITY	performance_alerts.severity	INFO, WARNING, CRITICAL
DIRECTION	driver_explainability.direction	POSITIVE, NEGATIVE, NEUTRAL

17. Changelog vs v2.0

Change	Description
New Layer J	Added Portfolio Construction layer with 4 tables: portfolio_plans, market_portfolio_scores, portfolio_capital_allocations, portfolio_return_metrics.
Expanded Layer A	Added assumption_lineage and governance_events to Layer A for full audit trail and approval workflow support.
Risk Framework	Formalized Layer H with 5 tables: risk_objects, risk_scenarios, monte_carlo_runs, monte_carlo_summaries, trigger_log.
Decision Support	Added decision_memory and decision_log to Layer I to support institutional memory and compliance audit.
Capital Strategy	Added capital_allocation_plans and dilution_models to Layer E for portfolio capital optimization.
Explainability	Added driver_explainability and variance_attribution to Layer G for 'why did KPI change?' analysis.
Confidence Scoring	Added assumption_confidence to Layer C for per-assumption confidence metadata.
Working Capital	Added working_capital_policies to Layer D (promoted from optional in v2 to Recommended v1.5).
Optional Extensions	Explicitly segregated 11 operational tables into Optional Extension layer with clear Phase 6 deferred status.
Audit Columns	Standardized AUDIT tail (company_id, created_at, updated_at, created_by, updated_by, deleted_at) applied to all 73 tables.
Total Tables	Grew from 37 tables (v2.0) to 73 tables (v3.0): 62 core/recommended/phase2-3 + 11 optional extension.